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GitHub Link	https://github.com/suri307/MLA0403-DEEP-LEARNING/tree/main/ASSESSMENTS/CO5/CO5-AT3

SIMATS ENGINEERING

CO5 - ASSESSMENT TOOL 3

Technical Presentation

COURSE NAME: Deep Learning **COURSE CODE: MLA04**
SLOT : **Faculty : Dr.Arul Raj A.M**

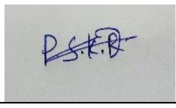
COURSE OUTCOME COVERED

CO5: Students will be able to analyze and explain advanced generative deep learning models including Autoencoders, Deep Belief Networks, Boltzmann Machines, Deep Boltzmann Machines and Generative Adversarial Networks. They will be able to compare their architectures, explain their learning mechanisms, and apply suitable generative models to real-world problems. (BTL-3)

Course Outcome Weightage: 35% **Duration: 90 minutes**
Assessment Tool Weightage: 50% **Mark : 25**

Code of Conduct:

I P Surendra Kumar Reddy (192425224) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:  **Name of the student:**

Criteria	Understanding of Problem Context (2)	Application of Concepts (3)	Problem-Solving Approach **(3)**	Accuracy of Solution **(2)**	Clarity ***(1)***	**Total (10)**
Weightage	***20 %***	***30%***	***30%***	***20 %***	***10 %***	**100 %**
Q1						
Total						

Signature of the Faculty: **Total Marks:**

PHOTO:

QUESTIONS:
3. Deep Belief Networks and Boltzmann Machines
Develop a technical presentation comparing Deep Belief Networks and Boltzmann Machines. Explain their architecture, energy-based learning, hidden and visible units, training process, and feature-learning capabilities. Illustrate their use in a suitable real-world application.

PRESENTATION ANSWER:

1. Introduction

Deep Belief Networks (DBNs) and Boltzmann Machines (BMs) are important **generative learning models** used in deep learning. They are designed to learn hidden patterns and relationships from data.

A Boltzmann Machine uses **visible and hidden units** and learns through an **energy-based approach**. A Restricted Boltzmann Machine (RBM) is a simplified form of a Boltzmann Machine that is easier to train. Multiple RBMs can be stacked to form a Deep Belief Network.

These models are useful for **feature learning, pattern recognition, recommendation systems, image processing, and dimensionality reduction**.

2. Problem Context

Modern applications generate large amounts of complex data such as images, speech, ratings, and sensor information. The important patterns in this data are not always directly visible.

Traditional machine learning methods may require manually selecting useful features. Deep learning attempts to overcome this limitation by automatically learning representations from raw data.

The main problem addressed by DBNs and BMs is:

How can a system automatically learn hidden patterns and meaningful features from complex data?

The models solve this problem by:

- Representing input data using visible units.
- Learning hidden relationships using hidden units.
- Assigning energy values to different data configurations.
- Learning useful features automatically.
- Building higher-level representations from lower-level features.

3. Relation to Deep Learning

Deep learning involves learning representations through multiple layers. DBNs are directly connected to this concept because they contain multiple layers that learn increasingly complex features.

The relationship can be represented as:

Boltzmann Machine → Restricted Boltzmann Machine → Multiple RBMs → Deep Belief Network

Connection with Deep Learning

- **Boltzmann Machine:** Introduces energy-based probabilistic learning.
- **RBM:** Provides a simpler and more efficient energy-based architecture.
- **DBN:** Stacks multiple learning layers to obtain hierarchical features.
- DBNs can learn representations without requiring all features to be manually designed.

For example:

Image Pixels → Edges → Shapes → Object Parts → Object Representation

4. Boltzmann Machine

A **Boltzmann Machine** is a stochastic, energy-based neural network used to learn the probability distribution of data.

It consists mainly of:

Visible Units

Visible units represent the **input data** provided to the network.

Examples:

- Image pixels
- User ratings
- Speech features
- Sensor values

Hidden Units

Hidden units represent **latent features** that are not directly present in the input.

For example, while processing an image, hidden units may learn:

- Edges
- Curves
- Shapes
- Patterns

Weights

Weights represent the strength of relationships between units.

Biases

Biases control the activation tendency of individual units.

5. Boltzmann Machine Architecture

A general Boltzmann Machine contains visible and hidden units with flexible connections.

Conceptual structure:

```
   Hidden Units
 H1   H2   H3   H4
 ↕    ↕    ↕    ↕
 ↕    ↕    ↕    ↕
 V1   V2   V3   V4
   Visible Units
```

In a standard Boltzmann Machine, connections can exist between many units. This makes the model powerful but computationally expensive to train.

6. Restricted Boltzmann Machine

A **Restricted Boltzmann Machine (RBM)** is a simplified version of a Boltzmann Machine.

It contains two layers:

1. Visible layer
2. Hidden layer

The important restriction is that there are:

- No connections between visible units.
- No connections between hidden units.
- Connections only between visible and hidden units.

Architecture

```
     Hidden Layer
  H1   H2   H3   H4
   |\  |\  /|  /|
   | \ | \/ | / |
   |  \| /\ |/  |
  V1   V2   V3   V4
     Visible Layer
```

The restricted structure makes RBMs easier and faster to train than general Boltzmann Machines.

7.Visible and Hidden Units

The roles of visible and hidden units can be understood using an image example.

Visible Layer

Suppose the input is an image.

The visible units receive the pixel values.

```
0  0  1  1
0  1  1  0
0  0  1  0
```

Hidden Layer

The hidden units can learn useful characteristics such as:

- Edges
- Curves
- Lines
- Corners
- Shapes

Therefore:

Input Pixels → Hidden Features → Useful Representation

This is called **feature learning**.

8. Energy-Based Learning

Energy-based learning is the central concept of Boltzmann Machines.

The model assigns an energy value to each possible configuration of visible and hidden units.

Basic Principle

Low Energy → More likely pattern

High Energy → Less likely pattern

During training, the model adjusts its weights so that training examples receive lower energy.

RBM Energy Function

$$E(v,h) = -\sum_i a_i\, v_i - \sum_j b_j\, h_j - \sum_{i,j} v_i\, w_{ij} h_j$$

Where:

- v_i= visible unit
- h_j= hidden unit
- w_{ij}= weight between visible and hidden units
- a_i= visible bias
- b_j= hidden bias

The energy function helps the model learn which combinations of visible and hidden units are more probable.

9. Training Process of RBM

The training process can be explained as follows:

Step 1: Initialize

Weights and biases are initialized, usually with small random values.

Step 2: Input Data

Training data is provided to the visible layer.

Step 3: Hidden Activation

The hidden units calculate their activation based on the visible input.

Step 4: Reconstruction

The hidden representation is used to reconstruct the visible data.

Step 5: Comparison

The original data is compared with the reconstructed data.

Step 6: Weight Update

Weights and biases are updated to improve the reconstruction and learned probability distribution.

Step 7: Repeat

The process is repeated for multiple training examples and iterations.

A common practical training method is **Contrastive Divergence (CD)**.

10. Deep Belief Network

A **Deep Belief Network (DBN)** is a deep generative model that contains multiple layers of learned representations.

A DBN can be constructed by stacking multiple RBM-like learning layers.

Architecture

```
  Hidden Layer 3
H31  H32  H33  H34
       ↓
  Hidden Layer 2
H21  H22  H23  H24
       ↓
  Hidden Layer 1
H11  H12  H13  H14
       ↓
    Input Data
```

Each layer learns a different level of representation.

11. Hierarchical Feature Learning

One of the major capabilities of DBNs is **hierarchical feature learning**.

Each layer learns features from the previous layer.

Example: Image Recognition

Raw Image
↓
Pixels
↓
Edges
↓
Shapes
↓
Object Parts
↓
Complete Object

For example:

- **First hidden layer:** learns edges and simple patterns.
- **Second hidden layer:** learns shapes and combinations of edges.
- **Higher layer:** learns complex structures.
- **Final representation:** captures meaningful information about the object.

Thus, DBNs automatically learn increasingly abstract features.

12. DBN Training Process

Traditional DBN training consists of two major stages.

Stage 1: Layer-Wise Unsupervised Pretraining

Each layer is trained separately.

Input Data
↓
Train RBM 1
↓
Learn Features
↓
Train RBM 2
↓
Learn Higher Features
↓
Train RBM 3

This allows the network to learn useful representations without requiring labels at every stage.

Stage 2: Fine-Tuning

After pretraining, the complete network can be fine-tuned for a particular task using available labeled data.

Examples:

- Classification
- Prediction
- Recognition

13. DBN vs Boltzmann Machine

Feature	**Boltzmann Machine**	**Deep Belief Network**
Type	Energy-based generative model	Deep generative model
Layers	Visible and hidden units	Multiple hidden layers
Connectivity	More general	Layered structure
Learning	Energy-based learning	Layer-wise pretraining
Feature learning	Hidden relationships	Hierarchical features
Complexity	Computationally expensive	More manageable
Main capability	Probability modeling	Deep representation learning

Key Difference

A Boltzmann Machine focuses on energy-based probabilistic modeling, while a Deep Belief Network uses multiple layers to learn hierarchical representations.

14. Feature-Learning Capabilities

Feature learning is an important reason for using these models.

Boltzmann Machine

Learns:

- Hidden dependencies
- Latent relationships
- Probability distributions

RBM

Learns:

- Compact representations
- Hidden patterns
- Useful latent features

DBN

Learns:

- Low-level features
- Mid-level features
- High-level features
- Hierarchical representations

For example:

Pixels → Edges → Shapes → Object

This reduces the need for manually designing every feature.

15. Real-World Application – Movie Recommendation System

A suitable real-world application of RBMs is a **movie recommendation system**.

Problem

A streaming service has information about:

- Users
- Movies
- Ratings
- Viewing history

However, the system does not know every user's hidden preferences.

The objective is:

Predict which movies a user is likely to prefer and recommend them.

16. Working of RBM in Recommendation

The recommendation process can be represented as:

User-Movie Ratings
↓
Visible Units
↓
Hidden Units
↓
Learn Hidden Preferences
↓
Predict Missing Preferences
↓
Generate Recommendations

Example

Suppose a user frequently watches:

- Interstellar
- Inception
- The Matrix
- Dune

The model may learn hidden preferences related to:

- Science fiction
- Futuristic themes
- Action
- Complex storytelling

The system can then predict that the user may like other movies with similar characteristics.

17. Problem-Solving Approach

The recommendation system can follow these steps:

1. **Identify the problem** – Recommend suitable movies to users.
2. **Collect data** – Gather user ratings and viewing history.
3. **Preprocess data** – Convert user-movie interactions into a suitable format.
4. **Train RBM** – Learn hidden user preferences.
5. **Extract features** – Identify latent preference patterns.
6. **Predict preferences** – Estimate ratings for unseen movies.
7. **Rank movies** – Arrange movies according to predicted preference.
8. **Generate recommendations** – Recommend the highest-ranked movies.
9. **Evaluate performance** – Measure prediction and recommendation quality.

18. Other Real-World Applications

1. Image Recognition

DBNs can learn visual features such as edges, shapes and object patterns.

2. Speech Recognition

The models can learn representations from speech signals.

3. Recommendation Systems

RBMs can learn hidden user preferences and predict missing ratings.

4. Anomaly Detection

Unusual patterns can be identified when their learned representation differs significantly from normal patterns.

5. Dimensionality Reduction

Complex high-dimensional data can be represented using a smaller set of learned features.

19. Advantages

Boltzmann Machines

- Can learn complex probability distributions.
- Can discover hidden relationships.
- Supports unsupervised learning.
- Useful for generative modeling.

Deep Belief Networks

- Learns hierarchical features automatically.
- Supports unsupervised pretraining.
- Can process complex and high-dimensional data.
- Reduces dependency on manually engineered features.
- Useful for representation learning.

20. Limitations

- General Boltzmann Machines are computationally expensive.
- Training can be slow.
- Energy-based optimization can be difficult.
- Performance depends on suitable hyperparameters.
- DBNs are less commonly used in modern applications compared with CNNs, Transformers and other advanced architectures.
- Large datasets may require significant computational resources.

21. Key Findings

The important findings from the comparison are:

- Boltzmann Machines use **energy-based probabilistic learning**.
- Visible units represent observed data.
- Hidden units learn latent features.
- RBMs simplify the architecture of Boltzmann Machines.
- DBNs can be constructed using multiple layers of learned representations.
- DBNs are capable of **hierarchical feature learning**.
- Contrastive Divergence is commonly used for practical RBM training.
- Recommendation systems demonstrate how hidden preferences can be learned from user behavior.

22. Conclusion

Deep Belief Networks and Boltzmann Machines are important models in the development of deep learning. Boltzmann Machines introduced the concept of **energy-based probabilistic learning**, while Restricted Boltzmann Machines provided a simpler structure for practical training.

Deep Belief Networks extend this concept by combining multiple layers to learn **hierarchical representations**. These models can automatically discover useful features from complex data.

Their application in recommendation systems demonstrates how hidden preferences can be learned from user behavior. Although modern deep learning architectures such as CNNs and Transformers are more widely used today, understanding BMs, RBMs and DBNs provides a strong foundation for understanding **generative models, latent representations and energy-based learning**.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand